

Supporting Materials: Multi-scale modeling of APC and β -catenin regulation in the human colonic crypt

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A Deriving Parameters for Normal Cell Motion

We seek parameters that describe normal crypt dynamics in the sense of cell movement and cell reproduction in the proliferative zone. To this end, we consider a model for cell density, $q(s, \tau)$, by coupling the effects of cell motion and proliferation from the previous sections. We obtain the following

$$\frac{\partial q}{\partial \tau} = \frac{\partial}{\partial s} \left(\frac{\alpha}{q^2} \frac{\partial q}{\partial s} \right) + H(\sigma - \sigma_p) \frac{\ln 2}{T_c} q \quad \text{for } (s, \tau) \in (0, L) \times (0, T] \quad (\text{A.1})$$

$$\frac{\partial q}{\partial s}(0, \tau) = 0 \quad \text{for } \tau \in (0, T] \quad (\text{A.2})$$

$$q(L, \tau) = 1 \quad \text{for } \tau \in (0, T] \quad (\text{A.3})$$

$$q(s, 0) = q_0(s) \quad \text{for } s \in (0, L). \quad (\text{A.4})$$

In the APC regulation model of Section 2, the level of stemness, σ , is a function of s , and σ_p corresponds to a specific point along the crypt axis, i.e., there exists a point $s_p \in [0, L]$ such that $\sigma_p = \sigma(s_p)$. Hence, the Heaviside representation of the proliferative zone is analogous to $H(s_p - s)$. Furthermore, specific changes in the parameters of the underlying ODE models of Emerick et. al. [1] correspond to shifts in the proliferative zone, which we study in Section 4.

To demonstrate and analyze the dynamics of the model, we first non-dimensionalize it and assume the proliferative zone is described by $H(s_p - s)$. Consider the following non-dimensional scalings

$$\hat{q} = \ell q, \quad \hat{\tau} = \frac{\ln 2}{T_c} \tau, \quad \hat{s} = \frac{s}{L}. \quad (\text{A.5})$$

Here, we recall that ℓ is the resting spring length, and will be set to the value $\ell = 1$. Substituting our scaled variables and dropping the hat notation for convenience gives the following dimensionless problem:

$$\frac{\partial q}{\partial \tau} = \frac{1}{\gamma} \frac{\partial}{\partial s} \left(\frac{1}{q^2} \frac{\partial q}{\partial s} \right) + H(s_p - s) q \quad \text{for } (s, \tau) \in (0, 1) \times (0, \hat{T}] \quad (\text{A.6})$$

$$\frac{\partial q}{\partial s}(0, \tau) = 0 \quad \text{for } \tau \in (0, \hat{T}] \quad (\text{A.7})$$

$$q(1, \tau) = 1 \quad \text{for } \tau \in (0, \hat{T}] \quad (\text{A.8})$$

$$q(s, 0) = q_0(s) \quad \text{for } s \in (0, 1). \quad (\text{A.9})$$

Here, the parameter γ is a measure of the ratio of the time scale of cell proliferation to the time scale of cell movement. This parameter is given by

$$\gamma = \frac{L^2 \ln 2}{\ell^2 \alpha T_c}. \quad (\text{A.10})$$

Figure 1 gives the numerical, steady state solution of Equations (A.6) – (A.9) for $q(s, \tau)$ and $v_s(s, \tau)$ (see Equation (12)) in dimensional units for small and large values of γ . For these simulations, we use $q_0 = 1$. From Equation (A.6), we can see that if $\gamma \gg 1$, cell proliferation dominates the evolution of the cell number density. Cells are produced inside the proliferative zone faster than they can move up and out of the crypt. Therefore, an accumulation of cells collects in the base, eventually leading to an exponential increase in the cell number density. Furthermore, cells collect in the proliferative region and although they cannot escape, the gradient of cell concentration creates a faster cell velocity by the time they exit the crypt. In contrast, if $\gamma \ll 1$, cell motion dominates and cells are sloughed away as soon as they are produced. As we can see, for small γ , the cell number density is at equilibrium throughout the crypt, indicating that each cell is exactly one cell apart. Also, the cells gain speed as they move from the proliferative zone and maintain a relatively fast velocity as they are extruded out of the crypt. In this case, the cell number density approaches 1 at each level of the crypt, which is the steady state solution of the PDE if we neglect the source term. Indeed, a similar analysis to Murray et. al. of the steady state version of Equation (A.6) leads to an upper bound on γ in which exponential cell growth does not occur. In fact, we can write

$$\gamma \leq \frac{1}{s_p(2 - s_p)} =: \gamma^*. \quad (\text{A.11})$$

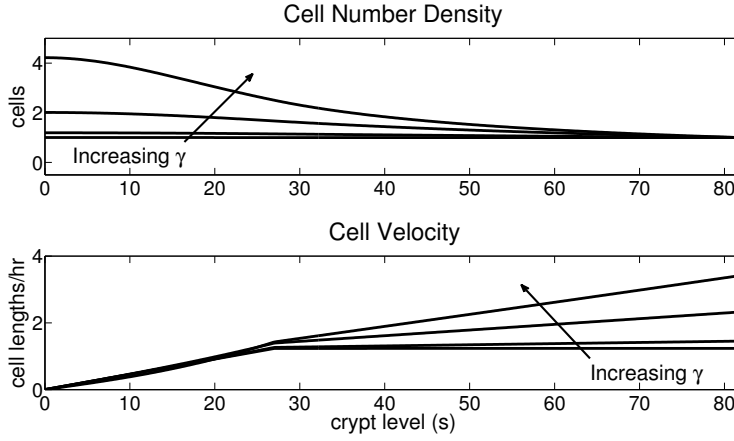


Fig. 1: Steady state solutions for cell number density, q , and cell velocity, v_s , of Equation A.6 for various γ . Parameters: $s_p = 27/L$, $T_c = 15.1954$, $L = 82$, and $\hat{T} = 50$.

We note that for $s_p = 27/L$, $\gamma^* = 1.81$ and any value above this threshold yields exponential growth of cell density. This upper bound ensures exponential growth will not occur, which we explore further in the next section.

The proliferative zone of the crypt contains both stem cells and transient amplifying (TA) cells. For simplicity, we lump these two populations together and denote them as simply proliferative cells that reside in the proliferative zone. It is known that stem cells are relatively immortal, dividing many times to produce a TA cell population while replenishing their own population [2]. Furthermore, stem cells have a very long mitotic cycle [3], taking approximately 100 hours to divide [4]. In contrast, a TA or rapidly proliferating cell divides almost 6 times faster, averaging anywhere near a 15-18 hour dividing rate [5]. Let N_s and N_t denote the number of stem and TA cells, respectively, in a normal homeostatic colonic crypt, and T_c be the average doubling time in the proliferative zone. Assuming the cell cycle time for stem and TA cells is T_s and T_t , respectively, and neglecting cell motion, there are approximately

$$N_s e^{\frac{\ln 2}{T_s} T_c} + N_t e^{\frac{\ln 2}{T_t} T_c} \quad (\text{A.12})$$

cells in the proliferative zone after T_c hours. To estimate T_c , we set

$$2(N_s + N_t) = N_s e^{\frac{\ln 2}{T_s} T_c} + N_t e^{\frac{\ln 2}{T_t} T_c}.$$

Typically there are about $N_s = 20$ stem cells and $N_t = 980$ TA cells in a normal crypt and, as noted before, we have $T_s = 100$ and $T_t = 15$ hours. Solving the above equation yields an average doubling time of a cell in the proliferative zone of $T_c \sim 15.1954$ hours.

Furthermore, we consider the non-dimensional Equation (A.6) to find relevant values for γ and s_p that yield accurate population totals in the entire crypt. A single crypt typically has between 3000 and 3500 cells, with about 1000 of those cells being proliferative. To find the total number of cells and the number of proliferative cells at any time τ , we compute

$$\begin{aligned} [\text{Total Cells}] &= 2\pi \int_0^L x(s) q(s, \tau) ds \\ [\text{Proliferative Cells}] &= 2\pi \int_0^{s_p} x(s) q(s, \tau) ds, \end{aligned}$$

where $x(s)$ is the radius of the crypt at level s , which is given by the parameterization, Equation (6). To obtain γ and s_p that correspond to the normal number of cells and proliferative cells, we numerically solve

Equations (A.6) – (A.9) for a set of possible parameter values $(s_p, \gamma) \in [0.05, 0.45] \times [0.05, 4.05]$. For each numerical run, we compute the steady state cell number density, q , and determine the total number of cells and total number of proliferative cells. Figure 2 shows the total number of cells and proliferative cells in (s_p, γ) space. The figure also shows acceptable contours – for total cell number, contours 3300, 3350, and 3400 are shown, and for total proliferative cells, contours 900, 1000, and 1100 are shown.

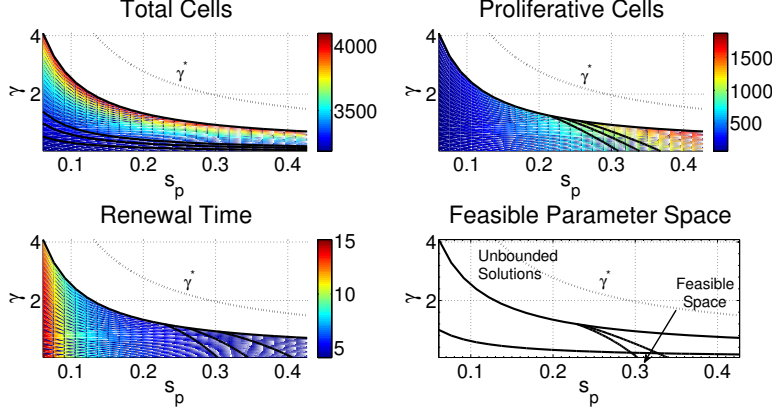


Fig. 2: Total cell count (top left), total proliferative cell count (top right), and renewal time (bottom left) in (γ, s_p) space. Contours for total cell count correspond to 3300, 3350, and 3400 cells; contours for proliferative cells correspond to 900, 1000, and 1100 cells; and contours for renewal time correspond to 4.5, 4.75, and 5 days. Optimal parameter region is bounded by the contours (bottom right).

Also shown in Figure 2 is the crypt renewal time plotted versus (γ, s_p) . The renewal time is defined as the time it takes for one cell to travel from the base of the crypt to the top. Typically, it takes between three and five days for the lining of the crypt to turn over completely. Renewal time for any τ is given by

$$[\text{Renewal Time}] = \int_1^L \frac{1}{v_s(s, \tau)} ds = - \int_1^L \frac{q^3}{\alpha} \left(\frac{\partial q}{\partial s} \right)^{-1} ds, \quad (\text{A.13})$$

where we integrate over the crypt starting at the second cell position (i.e., $s = 1$), since the velocity at $s = 0$ is zero. Contours shown in Figure 2 include 4.5, 4.75, and 5 days. Any point that is bounded inside the intersection of the three sets of contours is acceptable for a normal crypt. Figure 2 shows the optimal region in (γ, s_p) space that yields normal crypt dynamics. For our purposes, we let $\gamma = 0.1667$ and $s_p = 27/L$. We note here that the γ value for the small intestinal crypt is near $\gamma \sim 0.3$ [6], which is due to the shorter crypt size and overall lower cell count. All simulations are done using a finite difference scheme in space and using MATLAB's ODE solver `ode15s` to advance in time. Mesh refinement in the spatial direction, s , showed convergence of the numerical method.

B Numerical Scheme and Convergence

We consider the spatial discretization of the right hand side of Equation (25) using first order finite differences to approximate derivatives. Consider the rectangle $(s, \beta) \in [0, 1] \times [0, \beta_{\max}]$ where $\beta_{\max}$ is an upper bound on all non-dimensional β -catenin levels. Let the interval $[0, 1]$ be split into $N_s - 1$ subintervals with N_s equally spaced nodes, each a distance $ds = \frac{1}{N_s - 1}$ apart. Likewise, let the interval $[0, \beta_{\max}]$ be discretized

by N_β nodes, each a distance $d_\beta = \frac{\beta_{max}}{N_\beta - 1}$ apart. Now, we wish to find an approximation on the $N_\beta N_s$ distinct points denoted as

$$p(s, \beta, \tau) \approx \mathbf{P}_{i,j} = p(s_i, \beta_j, \tau), \quad (\text{B.1})$$

that holds for all τ and holds on each point $(s_i, \beta_j) = (0 + i d_s, 0 + j d_\beta)$, for $i = 1, 2, \dots, N_s$ and $j = 1, 2, \dots, N_\beta$. With this definition of $\mathbf{P}$, we also define the function $q(s, \tau)$ as a column vector

$$q(s, \tau) \approx \mathbf{q}_i = q(s_i, \tau) = \int_0^\infty p(s_i, \beta, \tau) d\beta, \quad (\text{B.2})$$

for each $i = 1, 2, \dots, N_s$. This integral is approximated on each β_j using the trapezoid rule in MATLAB. For future use, we extend this vector to a matrix that is unchanged across columns as

$$\mathbf{Q}_{i,j} = \underbrace{\begin{bmatrix} \mathbf{q}_i & \mathbf{q}_i & \dots & \mathbf{q}_i \end{bmatrix}}_{N_\beta \text{ columns}}. \quad (\text{B.3})$$

Furthermore, the function $v_\beta(s, \beta)$ is represented as the following matrix

$$v_\beta(s, \beta) \approx \mathbf{V}_{i,j} = v_\beta(s_i, \beta_j), \quad (\text{B.4})$$

where inside $v_\beta(s, \beta)$, we have the following column vector for the Wnt signal

$$W(s) \approx \mathbf{w}_i = W(y(s_i)), \quad (\text{B.5})$$

which is also expanded to the matrix constant in β as

$$\mathbf{W}_{i,j} = \underbrace{\begin{bmatrix} \mathbf{w}_i & \mathbf{w}_i & \dots & \mathbf{w}_i \end{bmatrix}}_{N_\beta \text{ columns}}. \quad (\text{B.6})$$

Finally, we denote the Heaviside function $H(\sigma - \sigma_p)$ as the following row vector

$$H(\sigma - \sigma_p) = \mathbf{h}_j = [0 \ 0 \ \dots \ 0 \ 1 \ 1 \ \dots \ 1], \quad (\text{B.7})$$

where the first non-zero entry occurs at the column j_p such that $\sigma_p = \sigma(\beta_{j_p})$. We can expand this to the following matrix

$$\mathbf{H}_{i,j} = \left\{ \begin{bmatrix} \mathbf{h}_j \\ \mathbf{h}_j \\ \vdots \\ \mathbf{h}_j \end{bmatrix} \right\}_{N_s \text{ rows}}. \quad (\text{B.8})$$

Therefore, there is a specific region in (s, β) space that consists of proliferative cells.

Spatial Operators in s : We seek appropriate differential matrices that represent each spatial operator in s -space and are second order accurate. We use the following method to approximate the derivative in s :

$$\frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] \approx \frac{1}{ds} \left[\left(\frac{1}{q^3} \frac{\partial q}{\partial s} p \right)_{i+\frac{1}{2}} - \left(\frac{1}{q^3} \frac{\partial q}{\partial s} p \right)_{i-\frac{1}{2}} \right] \quad (\text{B.9})$$

$$\approx \frac{1}{ds} \left[\frac{\mathbf{P}_{i+\frac{1}{2},j}}{\mathbf{Q}_{i+\frac{1}{2},j}^3} \left(\frac{\mathbf{Q}_{i+1,j} - \mathbf{Q}_{i,j}}{ds} \right) - \frac{\mathbf{P}_{i-\frac{1}{2},j}}{\mathbf{Q}_{i-\frac{1}{2},j}^3} \left(\frac{\mathbf{Q}_{i,j} - \mathbf{Q}_{i-1,j}}{ds} \right) \right] \quad (\text{B.10})$$

$$\approx \frac{1}{2ds} \left[\left(\frac{\mathbf{P}_{i+1,j}}{\mathbf{Q}_{i+1,j}^3} + \frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}^3} \right) \left(\frac{\mathbf{Q}_{i+1,j} - \mathbf{Q}_{i,j}}{ds} \right) - \left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}^3} + \frac{\mathbf{P}_{i-1,j}}{\mathbf{Q}_{i-1,j}^3} \right) \left(\frac{\mathbf{Q}_{i,j} - \mathbf{Q}_{i-1,j}}{ds} \right) \right], \quad (\text{B.11})$$

which holds for $i = 2, 3, \dots, N_s - 1$ and for all j . We now implement the boundary conditions using ghost points. For the condition at $s = 0$, we have $i = 1$. Hence, the boundary condition becomes

$$\frac{\partial q}{\partial s}(0, t) \approx \frac{\mathbf{Q}_{1,j} - \mathbf{Q}_{0,j}}{ds} = 0 \quad \Rightarrow \quad \mathbf{Q}_{0,j} = \mathbf{Q}_{1,j}. \quad (\text{B.12})$$

Furthermore, since there is no value for $P_{0,j}$, we'll use $P_{0,j} = P_{1,j}$ to average the p value when $s = 0$. Hence, the above spatial operator approximation for $i = 1$ is

$$i = 1 : \quad \frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] \approx \frac{1}{2ds} \left[\left(\frac{\mathbf{P}_{2,j}}{\mathbf{Q}_{2,j}^3} + \frac{\mathbf{P}_{1,j}}{\mathbf{Q}_{1,j}^3} \right) \left(\frac{\mathbf{Q}_{2,j} - \mathbf{Q}_{1,j}}{ds} \right) - \left(\frac{2\mathbf{P}_{1,j}}{\mathbf{Q}_{1,j}^3} \right) \left(\frac{0}{ds} \right) \mathbf{Q}_{1,j} \right], \quad (\text{B.13})$$

which holds for all j . Now, to implement the Dirichlet condition on q at $s = 1$, we simply replace the last element of $\mathbf{q}_i$ with 1. That is,

$$\mathbf{q}_i = \begin{bmatrix} \int_0^\infty p(s_1, \beta, t) d\beta \\ \int_0^\infty p(s_2, \beta, t) d\beta \\ \vdots \\ \int_0^\infty p(s_{N_s-1}, \beta, t) d\beta \\ 1 \end{bmatrix} \quad (\text{B.14})$$

This means that for the remainder of the discretization, the function q is given above, and our matrix $\mathbf{Q}_{i,j}$ is still defined as before with the last row updated. With the Dirichlet condition, we cannot solve for the ghost point, so we simply use a backward difference scheme instead of the forward difference for $i = N_s$. Similar to the previous case for $i = 1$, we use $P_{N_s,j}$ for $P_{N_s+1,j}$ in the averaging process. We obtain

$$i = N_s : \quad \frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] \approx \frac{1}{2ds} \left[\left(\frac{2\mathbf{P}_{N_s,j}}{\mathbf{Q}_{N_s,j}^3} \right) \left(\frac{\mathbf{Q}_{N_s,j} - \mathbf{Q}_{N_s-1,j}}{ds} \right) - \left(\frac{\mathbf{P}_{N_s,j}}{\mathbf{Q}_{N_s,j}^3} + \frac{\mathbf{P}_{N_s-1,j}}{\mathbf{Q}_{N_s-1,j}^3} \right) \left(\frac{\mathbf{Q}_{N_s,j} - \mathbf{Q}_{N_s-1,j}}{ds} \right) \right], \quad (\text{B.15})$$

Now, we compile this approximation using differential matrices with embedded boundary conditions. It follows that the differential operators in s -space can be approximated by the following

$$\frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] \approx \frac{1}{2ds^2} \left\{ \left[\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}^3} \right) \mathbf{A}_s^f \right] \cdot [\mathbf{Q}_{i,j} \mathbf{D}_s^f] - \left[\left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}^3} \right) \mathbf{A}_s^b \right] \cdot [\mathbf{Q}_{i,j} \mathbf{D}_s^b] \right\}, \quad (\text{B.16})$$

which holds for all $i = 1, 2, \dots, N_s$ and $j = 1, 2, \dots, N_\beta$. Here, we have kept the MATLAB convention and used right hand multiplication to denote finite differences in s even though the s -space is discretized as columns. Also, the $(\cdot)$ notation denotes element-wise multiplication. The matrices are given by

$$\mathbf{A}_s^f = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 0 & 1 & 1 \\ 0 & \cdots & 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{A}_s^b = \begin{bmatrix} 2 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 1 & 0 \\ 0 & \cdots & 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{D}_s^f = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 0 & -1 & 1 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}, \quad \mathbf{D}_s^b = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ -1 & 1 & 0 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 1 & 0 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

Spatial Operators in β : We employ a similar finite difference scheme for the derivative with respect to β . Now, we are working with differences in j , for $j = 1, 2, \dots, N_\beta$. We have

$$\frac{\partial}{\partial \beta} \left[v_\beta p - \eta \frac{\partial p}{\partial \beta} \right] \approx \frac{1}{d_\beta} \left[\left(v_\beta p - \eta \frac{\partial p}{\partial \beta} \right)_{i,j+\frac{1}{2}} - \left(v_\beta p - \eta \frac{\partial p}{\partial \beta} \right)_{i,j-\frac{1}{2}} \right] \quad (\text{B.17})$$

$$\approx \frac{1}{d_\beta} \left\{ \left[\mathbf{V}_{i,j+\frac{1}{2}} \cdot \mathbf{P}_{i,j+\frac{1}{2}} - \eta \left(\frac{\mathbf{P}_{i,j+1} - \mathbf{P}_{i,j}}{d_\beta} \right) \right] \right. \quad (\text{B.18})$$

$$\left. - \left[\mathbf{V}_{i,j-\frac{1}{2}} \cdot \mathbf{P}_{i,j-\frac{1}{2}} - \eta \left(\frac{\mathbf{P}_{i,j} - \mathbf{P}_{i,j-1}}{d_\beta} \right) \right] \right\} \quad (\text{B.19})$$

$$\approx \frac{1}{d_\beta} \left\{ \left[\left(\frac{\mathbf{V}_{i,j+1} \cdot \mathbf{P}_{i,j+1} + \mathbf{V}_{i,j} \cdot \mathbf{P}_{i,j}}{2} \right) - \eta \left(\frac{\mathbf{P}_{i,j+1} - \mathbf{P}_{i,j}}{d_\beta} \right) \right] \right. \quad (\text{B.20})$$

$$\left. - \left[\left(\frac{\mathbf{V}_{i,j} \cdot \mathbf{P}_{i,j} + \mathbf{V}_{i,j-1} \cdot \mathbf{P}_{i,j-1}}{2} \right) - \eta \left(\frac{\mathbf{P}_{i,j} - \mathbf{P}_{i,j-1}}{d_\beta} \right) \right] \right\},$$

which holds for all i and for $j = 2, 3, \dots, N_\beta - 1$. Now, we consider the no flux boundary conditions for when $j = 1$ and $j = N_\beta$. For $\beta = 0$, we have the following condition approximation

$$\begin{aligned} v_\beta(s, 0) p(s, 0, t) - \eta \frac{\partial p}{\partial \beta}(s, 0, t) &\approx \mathbf{V}_{i,1} \cdot \mathbf{P}_{i,1} - \eta \left(\frac{\mathbf{P}_{i,1} - \mathbf{P}_{i,0}}{d_\beta} \right) = 0 \\ \Rightarrow \quad \mathbf{P}_{i,0} &= \left(1 - \frac{d_\beta}{\eta} \mathbf{V}_{i,1} \right) \cdot \mathbf{P}_{i,1}. \end{aligned} \quad (\text{B.21})$$

We will use this as the ghost point for the backward difference scheme. Similarly, for $\beta = \beta_{\max}$, we have

$$\begin{aligned} v_\beta(s, \infty) p(s, \infty, t) - \eta \frac{\partial p}{\partial \beta}(s, \infty, t) &\approx \mathbf{V}_{i,N_\beta} \cdot \mathbf{P}_{i,N_\beta} - \eta \left(\frac{\mathbf{P}_{i,N_\beta+1} - \mathbf{P}_{i,N_\beta}}{d_\beta} \right) = 0 \\ \Rightarrow \quad \mathbf{P}_{i,N_\beta+1} &= \left(1 + \frac{d_\beta}{\eta} \mathbf{V}_{i,N_\beta} \right) \cdot \mathbf{P}_{i,N_\beta}. \end{aligned} \quad (\text{B.22})$$

This equation will be used for the ghost point for the forward difference equation above. Both of these conditions hold for all i . With each of these equations, we can find an explicit formula for the spatial operators at $\beta = 0$ and $\beta = \beta_{\max}$. As before, when averaging two values where one is unknown, we use the same value there. We have

$$\begin{aligned} j = 1: \quad \frac{\partial}{\partial \beta} \left[v_\beta p - \eta \frac{\partial p}{\partial \beta} \right] &\approx \frac{1}{d_\beta} \left\{ \left[\left(\frac{\mathbf{V}_{i,2} \cdot \mathbf{P}_{i,2} + \mathbf{V}_{i,1} \cdot \mathbf{P}_{i,1}}{2} \right) - \eta \left(\frac{\mathbf{P}_{i,2} - \mathbf{P}_{i,1}}{d_\beta} \right) \right] \right. \\ &\quad \left. - \left[\left(\frac{0}{2} \right) \mathbf{V}_{i,1} \cdot \mathbf{P}_{i,1} - \eta \left(\frac{0}{d_\beta} \right) \mathbf{P}_{i,1} \right] \right\}, \end{aligned} \quad (\text{B.23})$$

which holds for all i . Similarly, we have

$$\begin{aligned} j = N_\beta: \quad \frac{\partial}{\partial \beta} \left[v_\beta p - \eta \frac{\partial p}{\partial \beta} \right] &\approx \frac{1}{d_\beta} \left\{ \left[\left(\frac{0}{2} \right) \mathbf{V}_{i,N_\beta} \cdot \mathbf{P}_{i,N_\beta} - \eta \left(\frac{0}{d_\beta} \right) \mathbf{P}_{i,N_\beta} \right] \right. \\ &\quad \left. - \left[\left(\frac{\mathbf{V}_{i,N_\beta} \cdot \mathbf{P}_{i,N_\beta} + \mathbf{V}_{i,N_\beta-1} \cdot \mathbf{P}_{i,N_\beta-1}}{2} \right) \right. \right. \\ &\quad \left. \left. - \eta \left(\frac{\mathbf{P}_{i,N_\beta} - \mathbf{P}_{i,N_\beta-1}}{d_\beta} \right) \right] \right\}, \end{aligned} \quad (\text{B.24})$$

which holds for all i . Now, we can assemble the spatial operator in β into the following matrix equation which holds for all i and j . We have

$$\frac{\partial}{\partial \beta} \left[v_\beta p - \eta \frac{\partial p}{\partial \beta} \right] \approx \frac{1}{d_\beta} \left\{ \left[\frac{1}{2} \mathbf{A}_\beta^f (\mathbf{V}_{i,j} \cdot \mathbf{P}_{i,j}) - \frac{\eta}{d_\beta} \mathbf{D}_\beta^f \mathbf{P}_{i,j} \right] - \left[\frac{1}{2} \mathbf{A}_\beta^b (\mathbf{V}_{i,j} \cdot \mathbf{P}_{i,j}) - \frac{\eta}{d_\beta} \mathbf{D}_\beta^b \mathbf{P}_{i,j} \right] \right\}, \quad (\text{B.25})$$

where we have multiplied on the left hand side to keep the convention of MATLAB. The matrices are

$$\mathbf{A}_\beta^f = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 0 & 1 & 1 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_\beta^b = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 1 & 0 \\ 0 & \cdots & 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{D}_\beta^f = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 0 & -1 & 1 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{D}_\beta^b = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ -1 & 1 & 0 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 1 & 0 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

Production Term R: We can easily discretize the production term on the right hand side of the partial differential equation by replacing the continuous form with the values at the discrete points. We have the following approximation

$$H(\sigma - \sigma_p)p \approx \mathbf{H}_{i,j} \cdot \mathbf{P}_{i,j}, \quad (\text{B.26})$$

where $(\cdot)$ denotes element wise multiplication.

Vectorized Matrices: We wish to vectorize each concentration matrix and convert any spatial differential matrices into $N_\beta N_s \times N_\beta N_s$ matrices using the Kronecker matrix product. For each matrix above, denote the following as column vectors

$$\mathbf{p}_k = \text{vec}(\mathbf{P}_{i,j}), \quad \mathbf{q}_k = \text{vec}(\mathbf{Q}_{i,j}), \quad \mathbf{v}_k = \text{vec}(\mathbf{V}_{i,j}), \quad \mathbf{w}_k = \text{vec}(\mathbf{W}_{i,j}), \quad \mathbf{h}_k = \text{vec}(\mathbf{H}_{i,j}), \quad (\text{B.27})$$

for $k = 1, 2, \dots, N_\beta N_s$. The vec operation stacks the columns of each matrix into a single column. For any matrices A, B , and C , we have

$$\text{vec}(ABC) = [C^T \otimes A] \text{vec}(B), \quad (\text{B.28})$$

where $\otimes$ denotes the Kronecker matrix product. Therefore, for the spatial operators in s , we have

$$\frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] \approx \frac{1}{2ds^2} \left\{ \left[\mathcal{A}_s^f \left(\frac{\mathbf{p}_k}{q_k^3} \right) \right] \cdot [\mathcal{D}_s^f \mathbf{q}_k] - \left[\mathcal{A}_s^b \left(\frac{\mathbf{p}_k}{q_k^3} \right) \right] \cdot [\mathcal{D}_s^b \mathbf{q}_k] \right\}, \quad (\text{B.29})$$

where

$$\mathcal{A}_s^f = [(\mathbf{A}_s^f)^T \otimes I_\beta], \quad \mathcal{A}_s^b = [(\mathbf{A}_s^b)^T \otimes I_\beta], \quad \mathcal{D}_s^f = [(\mathbf{D}_s^f)^T \otimes I_\beta], \quad \mathcal{D}_s^b = [(\mathbf{D}_s^b)^T \otimes I_\beta], \quad (\text{B.30})$$

where I_β denotes the sparse identity matrix of size $N_\beta \times N_\beta$. Furthermore, the spatial operator in β -space is given by

$$\frac{\partial}{\partial \beta} \left[v_\beta p - \eta \frac{\partial p}{\partial \beta} \right] \approx \frac{1}{d_\beta} \left\{ \left[\frac{1}{2} \mathcal{A}_\beta^f (\mathbf{v}_k \cdot \mathbf{p}_k) - \frac{\eta}{d_\beta} \mathcal{D}_\beta^f \mathbf{p}_k \right] - \left[\frac{1}{2} \mathcal{A}_\beta^b (\mathbf{v}_k \cdot \mathbf{p}_k) - \frac{\eta}{d_\beta} \mathcal{D}_\beta^b \mathbf{p}_k \right] \right\}, \quad (\text{B.31})$$

where the Kronecker matrices are given by

$$\mathcal{A}_\beta^f = [I_s \otimes \mathbf{A}_\beta^f], \quad \mathcal{A}_\beta^b = [I_s \otimes \mathbf{A}_\beta^b], \quad \mathcal{D}_\beta^f = [I_s \otimes \mathbf{D}_\beta^f], \quad \mathcal{D}_\beta^b = [I_s \otimes \mathbf{D}_\beta^b]. \quad (\text{B.32})$$

Here, I_s denotes the sparse identity matrix of size $N_s \times N_s$. Finally, for the production term R , we simply have

$$H(\sigma - \sigma_p)p \approx \mathbf{h}_k \cdot \mathbf{p}_k. \quad (\text{B.33})$$

To sum up, the vectorized versions for the spatial discretization of the system given by Equations (25) – (30) becomes

$$\frac{\partial \mathbf{p}_k}{\partial \tau} = \frac{1}{2\gamma ds^2} \left\{ \left[\mathcal{A}_s^f \left(\frac{\mathbf{p}_k}{q_k^3} \right) \right] \cdot [\mathcal{D}_s^f \mathbf{q}_k] - \left[\mathcal{A}_s^b \left(\frac{\mathbf{p}_k}{q_k^3} \right) \right] \cdot [\mathcal{D}_s^b \mathbf{q}_k] \right\} - \frac{1}{\eta d_\beta} \left\{ \left[\frac{1}{2} \mathcal{A}_\beta^f (\mathbf{v}_k \cdot \mathbf{p}_k) - \frac{\eta}{d_\beta} \mathcal{D}_\beta^f \mathbf{p}_k \right] - \left[\frac{1}{2} \mathcal{A}_\beta^b (\mathbf{v}_k \cdot \mathbf{p}_k) - \frac{\eta}{d_\beta} \mathcal{D}_\beta^b \mathbf{p}_k \right] \right\} + \mathbf{h}_k \cdot \mathbf{p}_k, \quad (\text{B.34})$$

which holds for all $k = 1, 2, \dots, N_\beta N_s$ and for all τ . We are now ready to employ the method of lines and implement the right hand side of this equation into MATLAB's `ode15s`. We state that mesh refinement in the spatial directions showed convergence of the numerical method.

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